Assignment 1

Unit: FIT3152

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In this assignment, Generative AI (ChatGPT) has been used. It is used in 2 places shown in last page of appendix.

1. Descriptive Analysis

a) The dimension of data is 100000, 36 meaning of 100000 rows long and 36 columns wide. All of the columns are in integer data type except for the “Country” column, which is char data type. Except “year”, most of the numerical attributes here has a range of at most 15 and with a min of -5 and max of 10. The attributes with same prefix tend to have same range, for example prefix “IL” range from -5 to 4, prefix “ICQ” range from -5 to 1, prefix “ACT” range from -5 to 2, prefix “C” range from -5 to 4. Only prefix “POL” has different upper bound ranging from 3 to 10 but with same lower bound of -5. The year shows that this data is collected from 1981 to 2023. “Country” is the only non-numerical responses where it has a total of 58 countries.

There are missing values for each column except for “Wave”, “Country” and “Year”. In terms of column, “ACT” prefix variables each has around 20000+ data missing, “ACTSport” (26,208 missing), “ACTArtsEd” (24,584), “ACTProfessional” (23,987), “ACTHumanitarian” (24,226), and “ACTUnions” (23,577). It is also same for political variable, “PolScale” (29,280 missing), “PolExperts” (20,689), and “PolLeader” (18,719). “HardWork” is also another variable that has 22677 missing values. The most extreme case is “CEU” with 75,361 missing values because it is a country specific question. For the rest, the number of missing values is still bearable to not affect the overall dataset.

2. Focus country vs all other countries as a group (independent of time)

a)

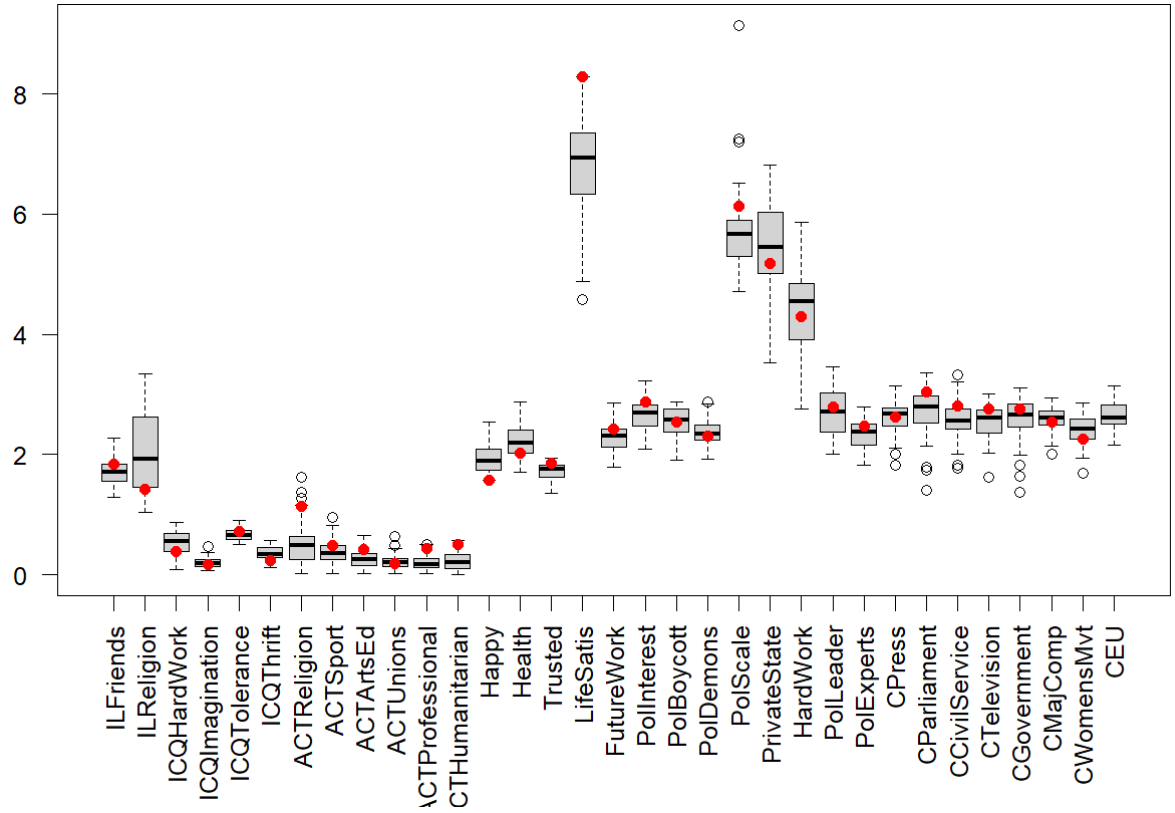


Diagram 2.1

Based on the taking the mean of each attribute for each country across the years after removing the nominal negative values to plot boxplot for each attribute. While the value for the focused country is pointed in red on the boxplot as shown in diagram 2.1.

By comparing Puerto Rico on other country, I noticed that Puerto Rico prioritise religion a lot according to “ILReligion” and “ACTReligion”, they fall near top 25% based on the level of importance for religion while being one of the countries having the most active membership of religious organisation. However, they see friends less important than other countries do where they fall near last 25%.

For child qualities, they prioritise hardwork and thrift saving money more than others countries, with both qualities being near top 25% but tolerance is less mentioned being the last 25% among countries. They also believe that hardwork can bring success than 50% of the countries.

Puerto Rico has high number of active memberships for most organisation compare to other country, including organisation in sport (ACTSport), Art (ACTArtsEd), professional (ACTProfessional), charity (ACTHumanitatrian) except for labour unions (ACTUnions).

However, people there still think that it's not good to have less importance placed at work (FutureWork) despite having low active membership of labour unions (ACTUnions) than other country.

Puerto Rico is a good country to live in because people there are one of the happiest countries (Happy) and healthier (Healthy) than near 75% of the surveyed country. They are also the country that is extremely satisfied about their life (LifeSatis).

In terms of politics on average, Puerto Rico is less interested in politics (PolInterest). They are more towards a democratic country where people prefer right side of politic scale (PolScale), prefer private ownership of business rather than government owned (PrivateState), slight disagree with strong leader without elections (PolLeader).

They are less confident with their government (CGovernment) including parliament (CParliament) and civil service (CCivilService) which all generally not confident compare to other country. They are only confident with some of the major companies (CMajComp) and the womens' movement organisation (CWomensMvt).

b)

Since we are predicting the confidence in social organisations, the attributes with "C" prefix will not be included in the predictors. The attributes are able to give some relationship but they are not good enough to predict confidence in social organisations in Puerto Rico. The attributes with "Pol" prefix generally seem to be the best group of predictors. Among the "Pol" prefix attributes, "PolScale" is one of the best predictors, where it is significant for all predictions. "PolInterest" is also another significant predictor for the top 3 predictable social organisations. Whereas the social organisations related to government can be relatively more reliably predicted is shown in order from "CGoverment", "CCivilServant" and "CParliament". These interpretations are based on the linear regression of all independent variables on each social organisation. For each organisation, only predictors with p-value smaller than 0.05 is considered significant to be display in table and explained below.

For CPress:

The multiple R square value is 0.083.

Independent variable, y	p-value	Significance Indicator
PolScale	2.03×10^{-2}	*
ILReligion	1.42×10^{-2}	*

For CParliament:

The multiple R square value is 0.135.

Independent variable, y	p-value	Significance Indicator
PolScale	1.43×10^{-2}	*
PolInterest	3.06×10^{-3}	**
ILReligion	3.01×10^{-3}	**
ACTReligion	1.37×10^{-2}	*
Trusted	4.56×10^{-2}	*

For CCivilService:

The multiple R square value is 0.153.

Independent variable, y	p-value	Significance Indicator
PolScale	1.57×10^{-3}	**
PolLeader	2.81×10^{-2}	*
PolInterest	2.88×10^{-4}	***
ICQHardWork	4.64×10^{-2}	*

For CTelevision:

The multiple R square value is 0.105.

Independent variable, y	p-value	Significance Indicator
PolScale	2.54×10^{-3}	**
ILFriends	2.12×10^{-2}	*
ILReligion	8.75×10^{-3}	**
ACTUnions	1.9×10^{-2}	*

For CGovernment:

The multiple R square value is 0.174.

Independent variable, y	p-value	Significance Indicator
PolScale	7.38×10^{-7}	***
PolInterest	7.06×10^{-7}	***
PolDemons	4.83×10^{-2}	*
ACTUnions	2.5×10^{-2}	*

For CMajComp:

The multiple square value is 0.116.

Independent variable, y	p-value	Significance Indicator
PolScale	2.28×10^{-2}	*
PrivateState	4.7×10^{-3}	**
ICQHardwork	2.42×10^{-2}	*
ICQTolerance	2.33×10^{-2}	*
Health	1.28×10^{-2}	*

For CWomensMvt:

The multiple square value is 0.06.

Independent variable, y	p-value	Significance Indicator
PolScale	1.12×10^{-2}	*
LifeSatis	1.55×10^{-2}	*

Overall Significance of Predictors:

Variables	CPress	CParliament	CCivilService	CTelevision	CGovernment	CMajComp	CWomensMvt	TotalSignificant
PolScale	*	*	**	**	***	*	*	11
ILReligion	*	**		**				5
PolInterest		**	***		***			8
ACTReligion		*						1
Trusted		*						1
PolLeader			*					1
ICQHardWork			*			*		2
ILFriends				*				1
ACTUnions				*	*			2
PosDemons					*			1
PrivateState						**		2
ICQTolerance						*		1
Health						*		1
LifeSatis							*	1

“CGovernment”, “CCivilService”, “CParliament” is arranged in the order of most relatively predictable to least predictable. This is based on the multiple R square value of each social

organisation with value of 0.174, 0.153 and 0.135 respectively. With higher multiple R square value, the more the variation of confidence in social organisation can be explained. Since “CGoverment” has highest relative multiple R square value of 0.174, it is the most reliable predicted because it’s variation about 17.4% is the most explained by the predictors.

In terms of predictors, “PolScale” is the best predictor because it is a significant predictor for all social organisation. “PolInterest” is also another good predictor because it has very small p-value for the top 3 social organisations meaning that it is a very significant predictor.

c)

For the prediction of “CPress” in other countries, the only 2 predictors for Puerto Rico not only became more significant, but an extra of 19 predictors also became significant. Within these significant predictors, “Trusted” is strongest predictors because it has p-value of smaller than 2 $x10^{-16}$ and largest magnitude of coefficient with an amount of 0.097. Some other predictors with similar p-value includes “PolInterest” and “ILReligion”. However, more significant predictors don’t explain the variation of “CPress” better, instead it became worse with a multiple R square value of 0.032.

With the same effect for the prediction of “CParliament” in other countries, the original 5 predictor for Puerto Rico not only became more significant, but also an extra of 17 predictors became significant. Within all attributes, the “Trusted” is still the strongest predictor because it still has p-value of smaller than 2 $x10^{-16}$ and largest magnitude of coefficient with amount of 0.186. Other predictors with similar p-value includes “PolScale”, “PolInterest”, “Hardwork”, “ILReligion” and “Happy”. The multiple R square value is 0.065 which the explaining of variation of “CParliament” also became worse.

For prediction of “CCivilService”, the original 4 predictors also became more significant with an extra of 13 new predictors. Among these attributes, “Trusted” is the strongest predictor with p-value of smaller than 2 $x10^{-16}$ and largest magnitude of coefficient with amount of 0.165. “PrivateState”, “PolInterest”, “Hardwork”, ILReligion”, “ILFriends” and “ACTUnions” also some predictors with similar p-value. The multiple R square is 0.047 which the explaining of variation of “CCivilService” became worse.

For prediction of “CTelevision”, the original 4 predictors became more significant with an extra of 16 new predictors. “Trusted” is the strongest predictor with p-value of smaller than 2 $x10^{-16}$ and largest magnitude of coefficient with amount of 8.431 x 10^{-2}. “PolScale”, “PolLeader”, “PrivateState”, “PolInterest”, “PolDemons”, “HardWork”, “ILReligion” and “ICQThrift” are also predictors with similar p-value. The multiple R square is 0.044 which the explaining of variation of “CTelevision” became worse.

For prediction of “CGovernment”, the original 4 predictors became more significant with an extra of 16 new predictors. “Trusted” is the strongest predictor with p-value of smaller than

2 $x10^{-16}$ and largest magnitude of coefficient with amount of 0.176. “PolScale”, “PolInterest”, “PolDemons”, “PolExpert”, “HardWork”, “ILReligion”, “Happy” and “LifeSatis” are also predictors with similar p-value. The multiple R square is 0.065 which the explaining of variation of “CGovernment” became worse.

For prediction of “CMajComp”, original 4 predictors became more significant with an extra of 17 new predictors. However, there is an unexpected case where an existing predictor “ICQTolerance” has no longer significant. “Trusted” is no longer the predictor with smallest p-value but still having largest coefficient of 0.070. The strongest predictors with smallest p-value are “PolScale”, “HardWork”, “ILReligion” and “Happy” with amount of smaller than 2 $x10^{-16}$. The multiple R square is 0.048 which the explaining of variation of “CMajComp” became worse.

For prediction of “CWomensMvt”, the original 2 predictors became more significant with an extra of 17 new predictors. “Trusted” is the no longer strongest predictor instead “PrivateState”, “ILReligion”, “ACTHumanitarian”, “Happy” and “LifeSatis” has the smallest p-value where it is smaller than 2 $x10^{-16}$. “ICQTolerance” has the largest magnitude of coefficient of 7.745 $x10^{-2}$. The multiple R square is 0.036 which the explaining of variation of “CWomensMvt” became worse.

In conclusion, the multiple R square of confidence in each social organisation for other countries is worse than the focus country group. The existing predictors in focus country generally become more significant in other countries. Moreover, the prediction for other countries has more predictor compare to focus country. In most case, the attribute “Trusted” is the strongest predictor in terms of smallest p-value and greatest magnitude of coefficient. The multiple R square for each social organisation in focus country has decreased when predicting other countries.

3. Focus country vs all other countries as a group (Over time)

a)

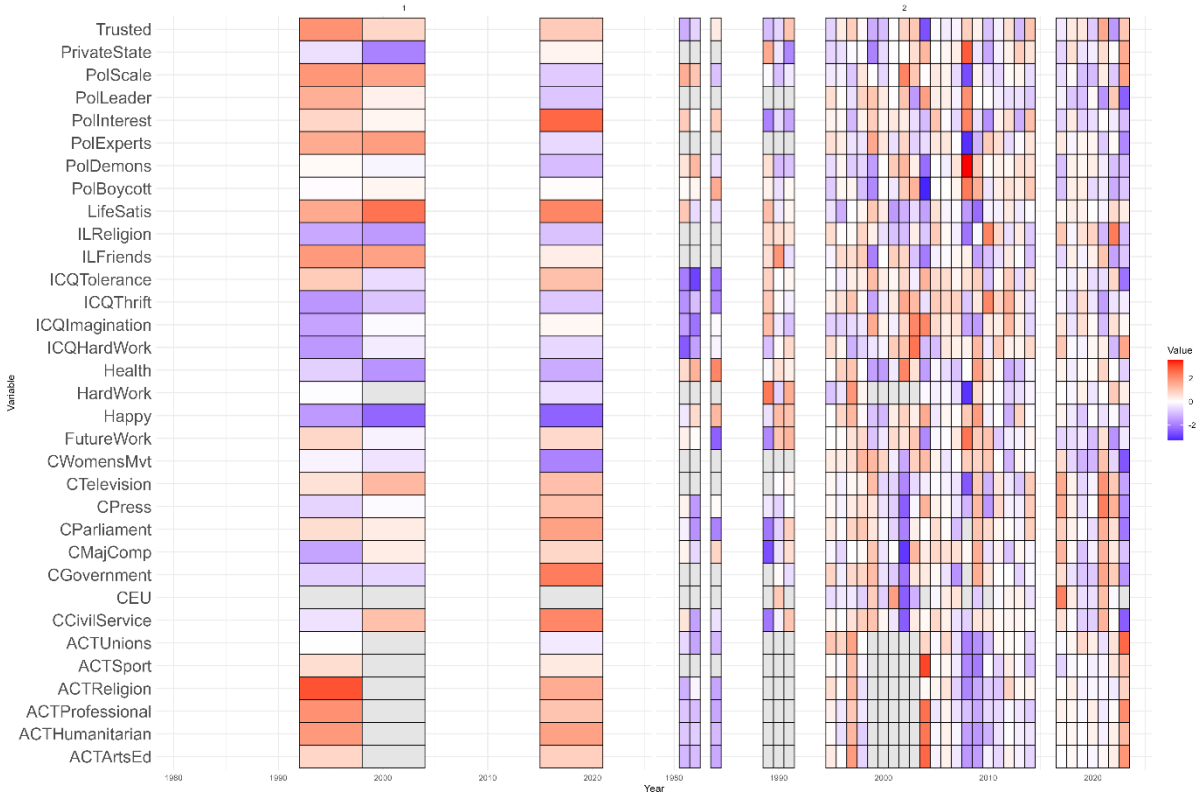


Diagram 3.1

In this question, the numerical data for each attribute is standardize into a range of -2 to 2 before creating the visualisation so that attributes with large range like “LifeSatis” doesn’t distort the visualisation of other attributes. The positive value shown in red after standardisation means above average, where the negative value shown in blue means below average, and zero shown in white indicates mean, whereas black or white without grid shows no data for that attribute. Each grid indicates a combination of year and attribute. Due to random sampling from the raw data, some years of data are missing, especially focusing country only has 3 years of data.

For the visualisation shown in diagram 3.1, there are 2 group of visualisations, left one is focusing country and right is other country. This visualisation will be explained in terms of group respectively, then compare and contrast.

For Puerto Rico, their trust is gained slowly throughout the year. Their private businesses are getting more and more then it either slowly taken over by government or government owned business had increased. From a political view, people prefer right side of political scale than gradually prefer left side. Meanwhile, they getting more interested in politics. People started from hate to like strong leader without elections and have experts making political decisions. They also started to join peaceful demonstrations while rarely join

boycotts throughout the years. In general, people quite satisfied with their life. Over the years, religion had always been important for them, while friends are getting more important too. For child qualities, tolerance is mentioned sometimes, imagination, thrift saving money and hard work are gradually getting noticed and felt important. They themselves also slowly thinks hard work can bring success. People there is generally happy and healthy over the years. They had always thought less importance placed in work is not a good thing. However, people started to loss their confidence in "The Press", parliament, government, major company, civil service and television. They only gained confidence for women movement organisation. European union is not recorded as Puerto Rico is not related. The memberships of most organisations have slight decrease over the year.

For other countries on average, people sometimes had trust but still slowly loss their trust. Businesses are private at first, then overtake by government business with sometimes reaching equilibrium and as for now private business has led. From a political view, people prefer right side of political scale at first and now, but often prefer left side in between. Meanwhile, they getting more interested in politics. People started from hate to like strong leader without elections and have experts making political decisions. They gradually take part in hosting boycott and also peaceful/ lawful demonstrations throughout the years. They are getting less satisfied of their life. People are not aware of importance of friends and religion, only sometimes they do and they slowly realise the importance. The trend is same for child qualities, where all qualities are not mentioned, but it is slowly discovered and prioritised. At first, people usually don't think they are healthy and happy, but these issues had been slowly resolved. They also gradually realise it is not good to have less importance placed in work. The confidence towards social organisation is great at start and now even though the confidence did drop in between. The number of active memberships in organisation is less at first and the number slowly increase across time.

These 2 groups generally have similarity and difference and it is shown in the table below.

Differences		
Puerto Rico	Attributes	Other Countries
Rise gradually	Trust	Starts high, dropped then recovered
Stable satisfied throughout	LifeSatis	Declining satisfaction
Consistently important	ILReligion & ILFriends	Delayed recognition
Directional, gradually right to left	PolScale	Cyclical, from right to left, then back to right
Rarely done	PolBoycott	Increasing participation
Slight decrease over time	All "ACT"	Increase over time

Similarities
Increase interest in politics over time
Become more accepting of strong leaders and expert decision-making
Increase peaceful/ lawful demonstrations
Increase in believing hard work leads to success
Increase in identification of importance of child qualities such as hard work, thrift, imagination
Fluctuation between private and government ownership, private government decrease relatively and increased

Heatmap is chosen as the chart to display the changes of variable over because it allows clear visualisation for each variable across the years. For large amount of variables, using heatmap will be more tidy and neat when no line crossing each other continuously over the years.

B)

In this questions, 3 graphics will be shown including the visualisation of ability of predictor across the years in focus country, visualisation of ability of predictor across the years in other countries and a comparison on both group for a certain social organisation. The ability in this question is defined by the relative importance of significant predictors with p value smaller than 0.05, where importance is defined by the p-value and estimates, with smaller p value and larger estimates, the more importance the predictor is and the higher its ranking. Notice that some predictors including “ACTReligion”, “ACTSport”, “ACTArtsEd”, “ACTUnions”, “ACTProfessional”, “ACTHumanitarian” and “HardWork” is not considered in the year of 2001 for both groups due to absence of data. The prediction for “CEU” in both group is also omitted due to more than half of the data is missing.

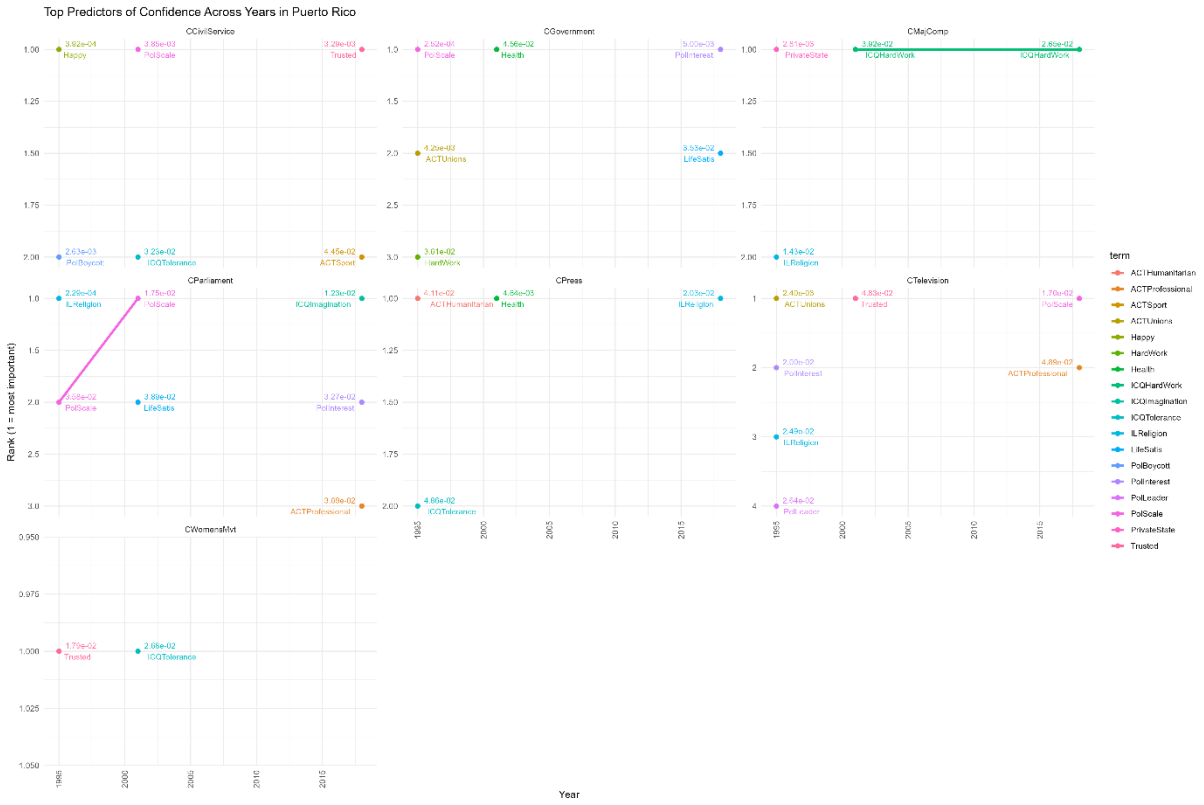


Diagram 3.2.1 Top Predictors of Confidence Across Years in Puerto Rico

The above diagram shows the ability of attributes to predict confidence in social organisations in others countries across 3 years, including 1995, 2001 and 2018 due to limited source of data.

For predicting “CCivilService”, the number of significant predictors is constant. This indicates that overall ability of important predictor has no change because when 2 predictors drop in ability, another 2 increased. In 1995, the “Happy” is the best predictors followed by “PolBoycott” the other important predictor, they are no longer significant in the coming years. In 2001, “PolScale” and “ICQTolerance” has risen in ability to be the important predictors but dropped in ability in 2018. In 2018, the ability of predictor in 2011 has dropped and replaced by “Trusted” and “ACTSport” with better ability to predict.

For predicting “CGovernment”, the number of significant predictors has decreased in overall. This indicates that the overall ability of important predictors has dropped because more predictors becoming less significance and less predictors become more significant. In “PolScale” is the best predictor followed by “ACTUnions” and “HardWork”, these predictors have dropped in ability in coming year and no longer significant. In 2001, “Health” is the best and the only predictor, it also lost its ability to predict in future. In 2018, “PolInterest” is the has risen in ability to be the best predictor and also “LifeSatis”.

For predicting “CMajComp”, the number of significant predictors has decreased in overall. This indicates that the overall ability of important predictors has dropped. Initially in 1995, “PrivateState” is the best predictor, followed by “ILReligion” but they did not maintain their ability in the coming years. In 2001, “ICQHardWork” had become the only predictor until 2018, its ability to predict has also slightly increase from 3.92×10^{-2} to 2.65×10^{-2} in terms of p value.

For predicting “CParliament”, the number of significant predictors has increased in overall. This indicates that the overall ability of important predictors has risen. In 1995, “ILReligion” and “PolScale” are the best predictors. After that “PolScale” has increased in ability to be thes best predictors in 2001, where “LifeSatis” also appear to be a new significant predictor. These predictors did not maintain its ability and all are replaced. In 2018, “ICQImagination” is the best predictor followed by “PolInterest” and “ACTProfessional”.

For predicting “CPress”, the number of significant predictors has decreased in overall. This indicates that the overall ability of important predictors has dropped. Initially in 1995, “ACTHumanitarian” and “ICQTolerance” are the significant predictor but they did not get to maintain their ability. “Health” is then the best predictor in 2001 and is replaced by “ILReligion” in 2018.

For predicting “CPress”, the number of significant predictors has decreased in overall. This indicates that the overall ability of important predictors has dropped. It’s start by having 4 predictors in 1995 including “ACTUnions”, “PolInterest”, “ILReligion” and “PolLeader”. None of these predictors get to maintain its ability and had dropped. In 2001, “Trusted” is the only predictor. It was replaced by “PolScale” and “ACTProfessional” in 2018.

For predicting “CPress”, the number of significant predictors has decreased. This indicates that the overall ability of important predictors has dropped. Initially, “Trusted” is the best predictor in 1995. It is then replaced by “ICQTolerance” in 2001. It is unexpected that none of the predictors get to increase in ability to be a significant predictor in 2018.

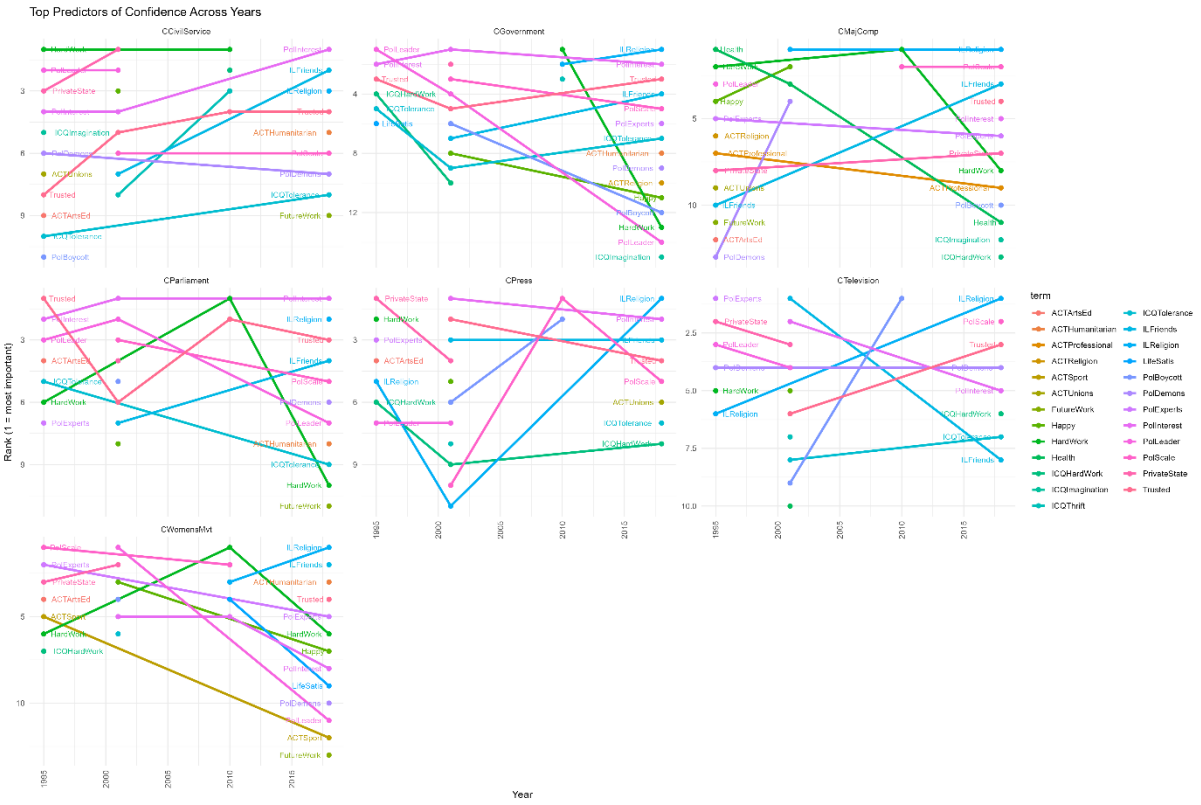


Diagram 3.2.2

The above diagram shows the ability of attributes to predict confidence in social organisations in others countries across 4 years, including 1995, 2001, 2010 and 2018 to ease visualisation and comparison with focus country. The amount is significant predictors is much more than the focus country group and the predictors get to retain its significance across years.

Starting from “CCivilService”, there are initially a lot of significant predictors, including “HardWork”, “PolLeader”, “PrivateState”, “PolInterest”, “ICQImaginaition”, “PolDemons”, “ACTUnions”, “Trusted”, “ACTArtEd”, “ICTolerance” and “PolBoycott”. At 1995, “HardWork”, “PolLeader” and “PrivateState” are the top 3 influential predictors of “CCivilService”. However, less than half of the predictors get to maintain its ability to predict across the years. Within these predictors that got remained, most of the predictors had increased in ability to predict, “PolInterest” has become the best predictor with significant rise of ranking from top 4 to top 1. Another apparent increase is “ILFriends” where it is not significant initially but starting from 2001, its ability to predict had increase until becoming the top 2 predictors in 2018. “Trusted” is also a predictor that has gradually increase in predictability, it was 9th best predictor in 1995, then rise to 5th, lastly in 4th place. “ICQTolerance” is also an has an observable rise from 10th place to 8th. Aside from rising ability, there are ability that has dropped like “PolDemons” from 6th place to 7th. In the latest predictions, there are few new significant predictors, there are “ILReligion”, “ACTHumanitarian” and “FutureWork”. Surprisingly, the best predictor, “HardWork” is no

longer consider as a predictor after 2010. Since the number of predictors has increased, the ability of predictors has increased including the most of predictors that are already significant.

For prediction in “CGovernment”, the number of significant predictors has increased over times. 6 predictors initial predictors had a clear decreased in ability of prediction, “PolLeader” was the best predictors but it continuously drops to 4th and 14th place. “ICQHardWork” had decreased and removed, while “ICQTolerance” with 5th place has a sharped drop and slight recover to 7th best predictor. Only a single predictor that appear in 2001 has increase in ability from 7th place to 4th place. Overall, the initial predictors from 1995 had decreased in ability and replaced by more other significant predictors such as “ILReligion” that suddenly appear as top 2 predictor in 2010 had become that best predictors lately. “Hardwork” that suddenly become best predictor in 2010 also had a significant drop at 2018. Since the number of significant predictors had increased over years, the overall ability of predictors had increased excluding the predictors that are already significant.

For prediction in “CMajComp”, the number of significant predictors is the same initially and in latest prediction with a big drop in between. There are 6 predictors gets to maintain its significance with 4 predictors had decreased in relative predictability and 2 increased. The others predictor had been removed and replace by other predictors across time. Among the drop, “Health” used to be the best predictors had drop significantly to 11th place in 2018. “HardWork” is again rising to be best predictors in 2010 then dropping 7 places to 8th. “PolExpects” and “ACTProfessional” has slight drop with 1 place and 2 place respectively. Generally, since the number of predictors is constant after years, this showed that the number of significant predictors removed are equally replaced, indicating the half of the other predictors has drop and another half predictors has increased, where the predictors that has always been significant has generally dropped in ability of predicting.

For prediction in “CParliament”, the number of predictors has increased over time. There are 5 predictors that retain its importance across the years. However, most of them drop in terms of importance when more predictors become significance. This includes “Trusted” which is initially the best predicted dropping significantly and slight recover to 3rd place in 2018. “PolLeader” shows slight increase from 3rd place to 2nd and dropped to 7th place. Whereas “ICQTolerance” dropped 4 places down. Similar to previous cases, “HardWork” had rose to be the best predictor in 2010 and drop sharply towards the 10th place. Only “PolInterest” gets to rise and become the best predictors in 2018. “ILFriends” which appears later also increase to 4th place. In general, the number of predictors has increased, indicating the ability of prediction for most predictors has improved except for most of the predictors that has been significant since 1995.

For prediction in “CPress”, the number of predictors has increased and slight recover over time. Only 2 predictors had retained it ability and importance across years, “ILReligion” had a dramatic change by a sudden drop of placement and sharp rise of ability to be the best predictor in 2018. Whereas “ICQHardWork” has dropped in general. There are also some predictors that become significant from 2001 which dropped in ability at 2018, this includes “PolInterest” from the best to 2nd, “Trusted” from 2nd to 4th, “ILFriends” has no changes, “PolScale” being the only attributes to rising sharply being the best predictor in 2010 from 10th then drop slightly to 5th order in 2018. Overall, by number of predictors, the prediction ability of most predictors has increased from 1995 to 2001 but decreased at 2018. Whereas predictors that exist across years, it shows decrease from 1995 to 2001 but half of it increase in 2018.

For prediction in “CTelevision”, the changes of number of predictors are similar to “CPress”. Among the 2 predictors that is important across years, only “ILReligion” has sharp increased to the best predictors for 2018. Where “PolDemon” remain no change in ranking. The top 3 predictors had decrease in ability and removed from the list of predictors in 2018. There are 4 predictors from 2001 that are included in 2018, where 2 decrease and 2 increase. Including “ILFriends” which was the best predictors has decreased significantly. “PolInterrest” also has dropped few places. “Trusted” and “ICQTolerance” has increase slightly in ability. In general, the ability of predictors has increased a lot in 2001 and recovered in 2018. Whereas predictors that exist across years, specific cases with sharp drop and rise cases is observed.

For prediction in “CWomensMvt”, a clear increase number of predictors are seen. Most of the predictors that are important across the years had decrease in ability to predict. There is a drastic drop for “PolLeader” being the best in 2001 but in 11th place in 2018. “PolExperts” also drop few places from 2nd to 5th place. Where “ACTSport” also decrease a lot from 2011 to 2018. Some decrease also observed in “Happy” and “PolInterest”. Other predictors like the best predictors for “PolScale” is no longer being the best predictors. There a lot of top predictors in 2018 with sudden rise in ability like “ILReligion”, “ILFriends”, “ACTHumanitarian” and “Trusted”. The usual phenomena of “HardWork” were also spotted here when it suddenly became the best predictor in 2010 and dropped. In general, most of the predictors has increased in ability because most has become significant, only the existing predictors has decreases after the appearance of other significant predictors.

There is no observation for predictions in “CEU” for Puerto Rico, so it is not discussed.

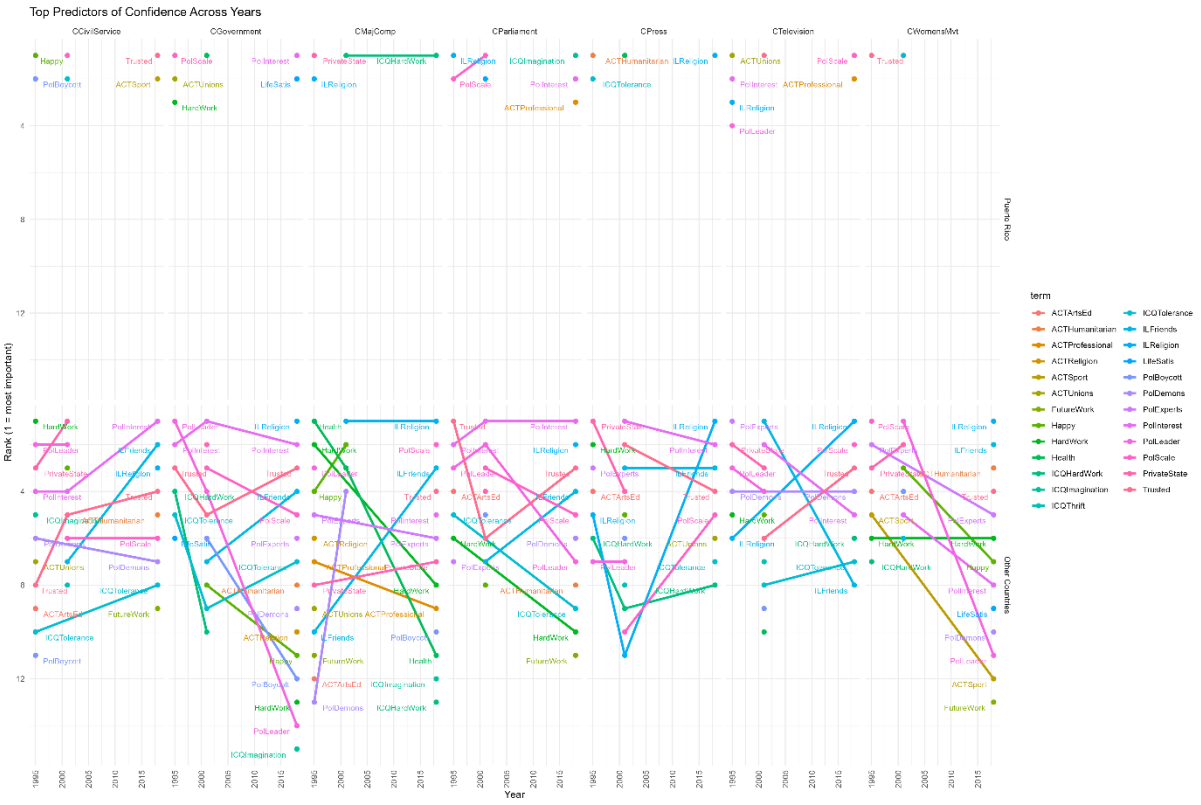


Diagram 3.2.3 Comparison of plot between 2 focus group

One of the major differences between 2 groups are the number of significant predictors. Predictors in Puerto Rico is often far less than the group of other countries. This shows that the regression model of Puerto Rico is way simpler than the one for other countries. Another key differences to point out is the volatility of predictors. The ability of predictors in Puerto Rico is much more stable, where less predictors get to rise into the list of significant predictors and drop out of it. These observations shows that the confidence in Puerto Rico is driven by a small and consistent set of factors. On the other hand, for other countries, their ranking within significant predictors changed a lot over time and they are pretty volatile over time. Observing the set of variables, Puerto Rico has a fixed set of dominant variables including “ILReligion”, “PolScale”, “Trusted” and “PolInterest”. However, for other countries wider set of variables can be seen, not in the ones in Puerto Rico but also includes “Happy”, “Health”, “HardWork”, “PolLeader”, “PolExperts”, some of the “ACT” variables have also beem repeatedly becoming the significant predictors.

In conclusion, Puerto Rico has a set of dominant and stable predictors, where others countries have wider variety and high volatility group of predictors.

The reason I chose bump chart to display the ability of variables in two groups is because variables are not always significant and important each year, the variables might not be display on chart in certain year, bump chart allows the sudden appearance and disappearance of predictors in each year compare to line chart. Moreover, bump chart can display the relative ability of predictors against other predictors using rank, this allows user

to understand the change of ability easier instead of using raw p value. Using rank also allows the y-axis to be less stretched, as compare to p-value that can range between 0.05 and smaller than $2x10^{-16}$.

Appendix

Data sampling:

```
set.seed(34216375)
VCData = read.csv("WVSExtract.csv")
VC = VCData[sample(1:nrow(VCData),100000, replace=FALSE),]
VC = VC[,c(1:3,sort(sample(4:50,25,replace=FALSE)),
sort(sample(51:65,8,replace=FALSE)))]
```

Question 1

```
dim(VC)
summary(VC)
str(VC)
#to replace missing value with NA
VCQ1 <- VC %>%
  mutate(across(where(is.numeric), ~ ifelse(. < 0, NA, .)))
#calculate all NA in each column
colSums(is.na(VCQ1))
```

Question 2a

```
library(dplyr)
VC1 <- VC %>%
  mutate(across(where(is.numeric), ~ ifelse(. < 0, NA, .))) %>%
  group_by(Country) %>%
  summarise(across(where(is.numeric), ~ mean(.x, na.rm = TRUE)))
View(VC1)
```

```
#ploting the box plot for question 2a

png("boxplot_PRI.png", width = 1200, height = 800, res = 150)


par(mar = c(7, 2, 0, 1))

#boxplot for overall dataset

boxplot(VC1[,4:36], las = 2)

#the red point to indicate data of focus country

PRI <- VC1[41,4:36]

points(1:length(PRI), PRI, col = "red", pch = 19)

dev.off()
```

Question 2b

```
VC2 <- VC %>%

  mutate(across(where(is.numeric), ~ ifelse(. < 0, NA, .))) %>%

  filter(Country == "PRI")


model <- lm(CPress ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)


model <- lm(CParliament ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)
```

```
model <- lm(CCivilService ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)

model <- lm(CTelevision ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)

model <- lm(CGovernment ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)

model <- lm(CMajComp ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)

model <- lm(CWomensMvt ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC2)

summary(model)
```

Question 2c

```
VC3 <- VC %>%

  mutate(across(where(is.numeric), ~ ifelse(. < 0, NA, .))) %>%
```

```
  filter(Country != "PRI")


model1 <- lm(CPress ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)

model1 <- lm(CParliament ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)

model1 <- lm(CCivilService ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)

model1 <- lm(CTelevision ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)

model1 <- lm(CGovernment ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
```

```
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)

model1 <- lm(CMajComp ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott +
PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)

model1 <- lm(CWomensMvt ~ PolScale + PolLeader + PrivateState +PolInterest + PolBoycott
+ PolDemons + PolExperts + HardWork +ILFriends + ILReligion + ICQHardWork +
ICQImagination + ICQTolerance + ICQThrift + ACTReligion + ACTSport + ACTArtsEd +
ACTUnions + ACTProfessional + ACTHumanitarian + Happy + Health + Trusted + LifeSatis +
FutureWork, data= VC3)


summary(model1)
```

Question 3a

```
VCGroup <- VC %>%
  mutate(Focus = ifelse(Country == "PRI", 1, 2))


VCQ3 <- VCGroup %>%
  mutate(across(-c(Year, Focus), ~ ifelse(. < 0, NA, .))) %>%
  group_by(Year, Focus) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE), .groups = "drop") %>%
#scale each column so it won't distort the colour scale in heat map
  mutate(across(-c(Year, Focus), ~ as.numeric(scale(.)))) %>%
  arrange(Year) %>%
  select(-Wave)
```

```
VCQ3Long <- VCQ3 %>%
  pivot_longer(
    cols = -c(Year, Focus),
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  mutate(Value = ifelse(is.nan(Value), NA, Value))


h <- ggplot(VCQ3Long, aes(x = Year, y = Variable, fill = Value)) +
#add grid to differentiate the each year and variable
  geom_tile(colour = "Black",  size=0.4) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", na.value = "grey90") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 18)) +
#separate and plot 2 group in 1 chart
  facet_wrap(~Focus)


ggsave("heat_map.png", plot = h,
       width = 18, height = 12, dpi = 300)
```

Question 3b

Graphic 1

```
library(dplyr)
install.packages("backports")
library(broom)


VCQ3b <- VCGroup %>%
```

```
  mutate(across(-c(Year, Focus, Country), ~ ifelse(. < 0, NA, .))) %>%
  mutate(across(-c(Year, Focus, Country), ~ as.numeric(scale(.)))) %>%
  filter(Country == "PRI") %>%
  select(-Wave, -CEU)

print(unique(VCQ3b$Year))

bump_data_focus <- data.frame()
predictors <- c("PolScale","PolLeader","PrivateState","PolInterest",
                "PolBoycott","PolDemons","PolExperts","ILFriends",
                "ILReligion","ICQHardWork","ICQImagination","ICQTolerance",
                "ICQThrift","Happy","Health","Trusted","LifeSatis",
                "FutureWork","ACTReligion","ACTSport","ACTArtsEd",
                "ACTUnions","ACTProfessional","ACTHumanitarian","HardWork")
predictors2001 <- c("PolScale","PolLeader","PrivateState","PolInterest",
                "PolBoycott","PolDemons","PolExperts","ILFriends",
                "ILReligion","ICQHardWork","ICQImagination","ICQTolerance",
                "ICQThrift","Happy","Health","Trusted","LifeSatis",
                "FutureWork")

confidence <- c("CPress", "CParliament", "CCivilService", "CTelevision", "CGovernment",
"CMajComp", "CWomensMvt")

for (c in confidence) {
  formula <- as.formula(
     paste(c, "~", paste(predictors, collapse = " + "))
    )
```

```
  lm1995 <- lm(formula, data=VCQ3b[VCQ3b$Year == 1995,])

  temp1995 <- tidy(lm1995) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 1995,
           estimate = abs(estimate))

  bump_data_focus <- bind_rows(bump_data_focus, temp1995)
}



for (c in confidence){
  formula <- as.formula(
     paste(c, "~", paste(predictors2001, collapse = " + "))
    )

  lm2001 <- lm(formula, data=VCQ3b[VCQ3b$Year == 2001,])

  temp2001 <- tidy(lm2001) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2001,
           estimate = abs(estimate))
  bump_data_focus <- bind_rows(bump_data_focus, temp2001)
```

```
}



for (c in confidence) {
  formula <- as.formula(
     paste(c, "~", paste(predictors, collapse = " + "))
    )

  lm2018 <- lm(formula, data=VCQ3b[VCQ3b$Year == 2018,])

  temp2018 <- tidy(lm2018) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2018,
           estimate = abs(estimate))
  bump_data_focus <- bind_rows(bump_data_focus, temp2018)
}

bump_data_focus <- bump_data_focus %>%
  group_by(confidence, year) %>%
  arrange(p.value, desc(estimate)) %>%
  mutate(rank = row_number())%>%
  ungroup()

label_start_mid <-bump_data_focus %>%
  group_by(confidence, term) %>%
  filter(year != 2018)
```

```
label_end <- bump_data_focus %>%
 group_by(confidence, term) %>%
 filter(year == 2018)

p <- ggplot(bump_data_focus,
      aes(x = year,
          y = rank,
          group = term,
          color = term)) +
 geom_line(linewidth = 1.2) +
 geom_point(size = 2) +
 scale_y_reverse() +
 facet_wrap(~ confidence, scales = "free_y") +
 labs(
  title = "Top Predictors of Confidence Across Years in Puerto Rico",
  x = "Year",
  y = "Rank (1 = most important)"
 ) +
 theme_minimal() +
 theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)) +
 geom_text(data = label_start_mid,
          aes(label = term),
          hjust = -0.2,
          size = 3,
          vjust=1.5,
          show.legend = FALSE) +
  geom_text(data = label_start_mid,
```

```
        aes(label = sprintf("%.2e", p.value)),
        hjust = -0.2,
        vjust= -0.5,
        size = 3,
        show.legend = FALSE) +
   geom_text(data = label_end,
        aes(label = sprintf("%.2e", p.value)),
        hjust = 1.2,
        vjust=-0.5,
        size = 3,
        show.legend = FALSE) +
  geom_text(data = label_end,
        aes(label = term),
        hjust =1.2,
        vjust=1.5,
        size = 3,
        show.legend = FALSE)



ggsave("bump_chart_focus.png", plot = p,
      width = 18, height = 12, dpi = 300)
```

Graphic 2

```
library(dplyr)
install.packages("backports")
library(broom)

VCQ3b <- VCGroup %>%
```

```
  mutate(across(-c(Year, Focus, Country), ~ ifelse(. < 0, NA, .))) %>%
  mutate(across(-c(Year, Focus, Country), ~ as.numeric(scale(.)))) %>%
  filter(Country != "PRI") %>%
  select(-Wave, -CEU)

print(unique(VCQ3b$Year))

bump_data_focus <- data.frame()
predictors <- c("PolScale","PolLeader","PrivateState","PolInterest",
                "PolBoycott","PolDemons","PolExperts","ILFriends",
                "ILReligion","ICQHardWork","ICQImagination","ICQTolerance",
                "ICQThrift","Happy","Health","Trusted","LifeSatis",
                "FutureWork","ACTReligion","ACTSport","ACTArtsEd",
                "ACTUnions","ACTProfessional","ACTHumanitarian","HardWork")
predictors2001 <- c("PolScale","PolLeader","PrivateState","PolInterest",
                "PolBoycott","PolDemons","PolExperts","ILFriends",
                "ILReligion","ICQHardWork","ICQImagination","ICQTolerance",
                "ICQThrift","Happy","Health","Trusted","LifeSatis",
                "FutureWork")

confidence <- c("CPress", "CParliament", "CCivilService", "CTelevision", "CGovernment",
"CMajComp", "CWomensMvt")

for (c in confidence) {
  formula <- as.formula(
     paste(c, "~", paste(predictors, collapse = " + "))
    )
```

```
  lm1995 <- lm(formula, data=VCQ3b[VCQ3b$Year == 1995,])

  temp1995 <- tidy(lm1995) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 1995,
           estimate = abs(estimate))

  bump_data_focus <- bind_rows(bump_data_focus, temp1995)
}



for (c in confidence){
  formula <- as.formula(
     paste(c, "~", paste(predictors2001, collapse = " + "))
    )

  lm2001 <- lm(formula, data=VCQ3b[VCQ3b$Year == 2001,])

  temp2001 <- tidy(lm2001) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2001,
           estimate = abs(estimate))
  bump_data_focus <- bind_rows(bump_data_focus, temp2001)
```

```
}

for (c in confidence) {
  formula <- as.formula(
     paste(c, "~", paste(predictors, collapse = " + "))
    )

  lm2010 <- lm(formula, data=VCQ3b[VCQ3b$Year == 2010,])

  temp2010 <- tidy(lm2010) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2010,
           estimate = abs(estimate))
  bump_data_focus <- bind_rows(bump_data_focus, temp2010)
}


for (c in confidence) {
  formula <- as.formula(
     paste(c, "~", paste(predictors, collapse = " + "))
    )

  lm2018 <- lm(formula, data=VCQ3b[VCQ3b$Year == 2018,])

  temp2018 <- tidy(lm2018) %>%
    filter(term!= "(Intercept)") %>%
```

```
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2018,
           estimate = abs(estimate))
  bump_data_focus <- bind_rows(bump_data_focus, temp2018)
}

bump_data_focus <- bump_data_focus %>%
  group_by(confidence, year) %>%
  arrange(p.value, desc(estimate)) %>%
  mutate(rank = row_number())%>%
  ungroup()

label_start <- bump_data_focus %>%
  group_by(confidence, term) %>%
  filter(year == 1995))

label_end <- bump_data_focus %>%
  group_by(confidence, term) %>%
  filter(year == 2018)

 p <- ggplot(bump_data_focus,
       aes(x = year,
           y = rank,
           group = term,
           color = term)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
```

```
  scale_y_reverse() +
  facet_wrap(~ confidence, scales = "free_y") +
  labs(
    title = "Top Predictors of Confidence Across Years in Puerto Rico",
    x = "Year",
    y = "Rank (1 = most important)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)) +
  geom_text(data = label_start,
            aes(label = term),
            hjust = -0.2,
            size = 3,
            vjust=1.5,
            show.legend = FALSE) +
  geom_text(data = label_end,
            aes(label = term),
            hjust =1.2,
            vjust=1.5,
            size = 3,
            show.legend = FALSE)

ggsave("bump_chart_others.png", plot = p,
       width = 18, height = 12, dpi = 300)
```

Graphic 3

```
VCQ3b1 <- VCGroup %>%
  mutate(across(-c(Year, Focus, Country), ~ ifelse(. < 0, NA, .))) %>%
```

```
  mutate(across(-c(Year, Focus, Country), ~ as.numeric(scale(.)))) %>%
  filter(Country == "PRI") %>%
  select(-Wave, -CEU)


VCQ3b2 <- VCGroup %>%
  mutate(across(-c(Year, Focus, Country), ~ ifelse(. < 0, NA, .))) %>%
  mutate(across(-c(Year, Focus, Country), ~ as.numeric(scale(.)))) %>%
  filter(Country != "PRI") %>%
  select(-Wave, -CEU)
bump_data_focus <- data.frame()
predictors <- c("PolScale","PolLeader","PrivateState","PolInterest",
                "PolBoycott","PolDemons","PolExperts","ILFriends",
                "ILReligion","ICQHardWork","ICQImagination","ICQTolerance",
                "ICQThrift","Happy","Health","Trusted","LifeSatis",
                "FutureWork","ACTReligion","ACTSport","ACTArtsEd",
                "ACTUnions","ACTProfessional","ACTHumanitarian","HardWork")
predictors2001 <- c("PolScale","PolLeader","PrivateState","PolInterest",
                "PolBoycott","PolDemons","PolExperts","ILFriends",
                "ILReligion","ICQHardWork","ICQImagination","ICQTolerance",
                "ICQThrift","Happy","Health","Trusted","LifeSatis",
                "FutureWork")


confidence <- c("CPress", "CParliament", "CCivilService", "CTelevision", "CGovernment",
"CMajComp", "CWomensMvt")


for (c in confidence) {
  formula <- as.formula(
```

```
    paste(c, "~", paste(predictors, collapse = " + "))
  )

 lm1995 <- lm(formula, data=VCQ3b1[VCQ3b1$Year == 1995,])

 temp1995 <- tidy(lm1995) %>%
  filter(term!= "(Intercept)") %>%
  filter(p.value < 0.05) %>%
  mutate(confidence = c,
         year = 1995,
         estimate = abs(estimate),
         focus = 1)

 bump_data_focus <- bind_rows(bump_data_focus, temp1995)
}

for (c in confidence) {
 formula <- as.formula(
    paste(c, "~", paste(predictors, collapse = " + "))
  )

 lm1995 <- lm(formula, data=VCQ3b2[VCQ3b2$Year == 1995,])

 temp1995 <- tidy(lm1995) %>%
  filter(term!= "(Intercept)") %>%
  filter(p.value < 0.05) %>%
  mutate(confidence = c,
         year = 1995,
```

```
        estimate = abs(estimate),
        focus = 2)

  bump_data_focus <- bind_rows(bump_data_focus, temp1995)
}



for (c in confidence){
  formula <- as.formula(
    paste(c, "~", paste(predictors2001, collapse = " + "))
   )

  lm2001 <- lm(formula, data=VCQ3b1[VCQ3b1$Year == 2001,])

  temp2001 <- tidy(lm2001) %>%
   filter(term!= "(Intercept)") %>%
   filter(p.value < 0.05) %>%
   mutate(confidence = c,
        year = 2001,
        estimate = abs(estimate),
        focus = 1)
  bump_data_focus <- bind_rows(bump_data_focus, temp2001)
}

for (c in confidence){
  formula <- as.formula(
    paste(c, "~", paste(predictors2001, collapse = " + "))
```

```
    )

  lm2001 <- lm(formula, data=VCQ3b2[VCQ3b2$Year == 2001,])

  temp2001 <- tidy(lm2001) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2001,
           estimate = abs(estimate),
           focus = 2)
  bump_data_focus <- bind_rows(bump_data_focus, temp2001)
}



for (c in confidence) {
  formula <- as.formula(
      paste(c, "~", paste(predictors, collapse = " + "))
    )

  lm2018 <- lm(formula, data=VCQ3b1[VCQ3b1$Year == 2018,])

  temp2018 <- tidy(lm2018) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
           year = 2018,
```

```
        estimate = abs(estimate),
        focus = 1)
  bump_data_focus <- bind_rows(bump_data_focus, temp2018)
}


for (c in confidence) {
  formula <- as.formula(
     paste(c, "~", paste(predictors, collapse = " + "))
    )

  lm2018 <- lm(formula, data=VCQ3b2[VCQ3b2$Year == 2018,])

  temp2018 <- tidy(lm2018) %>%
    filter(term!= "(Intercept)") %>%
    filter(p.value < 0.05) %>%
    mutate(confidence = c,
        year = 2018,
        estimate = abs(estimate),
        focus = 2)
  bump_data_focus <- bind_rows(bump_data_focus, temp2018)
}
bump_data_focus <- bump_data_focus %>%
  group_by(focus, confidence, year) %>%
  arrange(p.value, desc(estimate)) %>%
  mutate(rank = row_number())%>%
  ungroup()
```

```
label_start <- bump_data_focus %>%
  group_by(focus, confidence, term) %>%
  filter(year == 1995)

label_end <- bump_data_focus %>%
  group_by(focus, confidence, term) %>%
  filter(year == 2018)

p <- ggplot(bump_data_focus,
       aes(x = year,
           y = rank,
           group = term,
           color = term)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  scale_y_reverse() +
  facet_grid(focus ~ confidence,
           labeller = labeller(focus = c("1" = "Puerto Rico",
                                         "2" = "Other Countries")))+
  labs(
    title = "Top Predictors of Confidence Across Years",
    x = "Year",
    y = "Rank (1 = most important)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)) +
  geom_text(data = label_start,
            aes(label = term),
```

```
        hjust = -0.2,
        size = 3,
        vjust=1.5,
        show.legend = FALSE) +
  geom_text(data = label_end,
        aes(label = term),
        hjust =1.2,
        vjust=1.5,
        size = 3,
        show.legend = FALSE)

ggsave("bump_chart_both.png", plot = p,
     width = 18, height = 12, dpi = 300)
```

(ChatGPT) AI declaration:

1. Method of saving box plot as .png in question 2a
    a. Prompt: How to save box plot () in .png
    b. It shows that the dev.off () is require to allow saving without errors
    c. The use of code fragment is shown in bold text

```
        png("boxplot_PRI.png", width = 1200, height = 800, res = 150)


        par(mar = c(7, 2, 0, 1))
        boxplot(VC1[,4:36], las = 2)
        PRI <- VC1[41,4:36]
        points(1:length(PRI), PRI, col = "red", pch = 19)
```

dev.off() # Added after guidance of chatgpt

2. Method of doing linear regression in a for loop in question 3b
    a. Prompt: How to use linear regression in for loop

b. The as.formula () function is given to allow the dependent variable to change according to for loop
c. Paste () allows multiple independent variables to fit inside function
d. The use of code fragment is shown in bold text

```
for (c in confidence) {
#added after guidance of chatgpt
 formula <- as.formula(
    paste(c, "~", paste(predictors, collapse = " + "))
  )

 lm1995 <- lm(formula, data=VCQ3b1[VCQ3b1$Year == 1995,])

 temp1995 <- tidy(lm1995) %>%
  filter(term!= "(Intercept)") %>%
  filter(p.value < 0.05) %>%
  mutate(confidence = c,
         year = 1995,
         estimate = abs(estimate),
         focus = 1)

 bump_data_focus <- bind_rows(bump_data_focus, temp1995)
}
```